

When Do Attention Sinks Emerge in VLM Pretraining — and Do Their Signatures Decouple?

From-scratch attention-sink emergence study on a 222M vision-language model · pipeline-validation + early-window stage · single-author, seeking mentor / corresponding author

THE QUESTION

Trained LLMs develop an **attention sink**: one token (usually the first) absorbs a large share of attention. In practice this co-occurs with two *representation-level* signatures at that token — **massive activations** (a handful of hidden dimensions with outsized norm) and **value-norm drain**. The literature treats the three as one phenomenon and studies it *at inference, on frozen models*. Two things are unmeasured: (1) **when/where** these form during *multimodal* pretraining, and (2) whether attention-concentration and the norm/representation signatures actually move together — or **decouple**. The first token here is a *SigLIP image patch*, not a BOS token, which makes the VLM setting a clean natural experiment.

SETUP

nanoVLM-222M, trained **from scratch**: a bidirectional vision **encoder** (SigLIP-B/16) whose patch embeddings are projected into the prefix of an autoregressive **decoder** LM (SmolLM2-135M architecture, 30 causal-attention layers). **The attention sinks we study live in that decoder's causal self-attention** — position 0 is the first image-patch token in its sequence — so this is fundamentally a decoder phenomenon, with the encoder supplying the visual prefix. Dense probing every 100 steps via an eager fp32 re-walk validated against the live forward (rel-err < 1e-2). Three signatures logged **separately** per (layer,head): concentration Sink^{0.3}, value-norm ratio, hidden-norm ratio. Four arms each isolate one lever (softmax baseline · Qiu σ-gate · sigmoid-without-softmax · text-pretrained init).

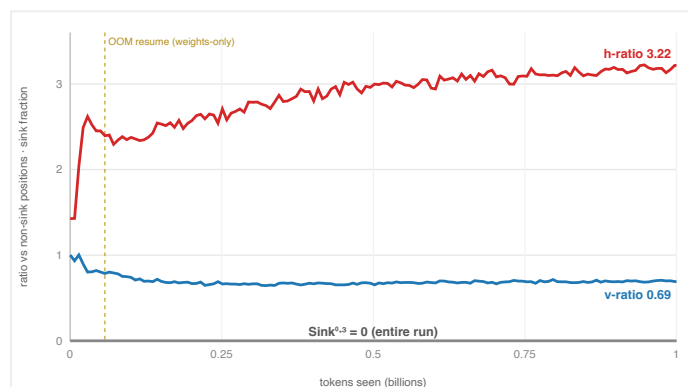


Fig 1 — Decoupling in the 1B-token from-scratch baseline. The concentration sink **never fires** (Sink^{0.3} = 0 the entire run) while the massive-activation ratio climbs **+130%** (1.43→3.22). Concentration and massive activation are dissociated within a single run. Dashed line = an OOM-driven weights-only restart, audited not to affect the signature trajectory.

WHY THIS SHOULD INTEREST YOU (EMBEDDINGS)

Massive activations are the same **outlier** / **"rogue" dimensions** that dominate embedding geometry and distort cosine similarity. The result here says attention-concentration and these outlier-dimension signatures can be **dialed independently** — i.e. a training-time lever on representation geometry, not merely on where attention points. Whether that identification (massive activation ⇒ embedding outlier dimension) is the right bridge is exactly one place I want your read.

QUESTIONS FOR YOU

- **Strength of the negative claim.** "The concentration sink never forms" currently rests on a **222M-parameter** model trained to **1B tokens** (single seed for this run), vs Gu et al.'s 1B-parameter / 5B-token text-LM setting. What's the minimum scale / seed count a reviewer needs to trust a *negative* result?
- **Framing & novelty.** Lead with decoupling, with emergence-timing, or with the representation-geometry bridge? Scoop risk is currently judged LOW–MEDIUM — does that match what you see in the field?
- **The outlier-dimension link.** Are the decoder's massive-activation dimensions the same **outlier** / **rogue dimensions** that dominate text-embedding geometry — same indices, same effect on cosine similarity / anisotropy? Is nailing that correspondence the contribution worth headlining, and what would you measure for it?

Status: 222M scale, ≤1B tokens, dense-probed; results above are real runs, not projections. This is the early-window / pipeline-validation stage, not the full scaled map. Code + raw per-step logs available on request.

	concentration Sink ^{0.3}	value-norm liv/l pos0/rest	massive-act. norm pos0/rest	val loss
baseline softmax / random LM	none 0.00	drained 0.71	moderate 2.2	1.161
g1gate σ-gate (Qiu)	weak 0.004	mild 0.82	suppressed 1.6	1.138
sigmoid no softmax (Gu)	strong 0.83	amplified 1.48	none 1.1	1.108
textinit pretrained text LM	total 0.85	severe 0.38	extreme 42	0.832

Fig 2 — Four-way dissociation. Holding data, optimizer and architecture fixed, each lever yields a *different* combination of the three signatures — e.g. **sigmoid** gives strong concentration with *no* massive activation; **text-init** inherits all three at step 0, relocated onto the first image token. They are independently controllable, not facets of one thing.